Isogeny-class trio: three 41–64-bit roots for a whole-class experiment

Date: 2026-10-17. Protocol: `PROTOCOL.md`, frozen before the B and P searches. PDF: `REPORT.pdf`. Data: `data/`. Tool: `selector/` (native Rust, no dependencies) and `identity/` (the repository's native ICV1 producer).

Question

The user asked for three curves, one binary, one Koblitz and one prime-field, between 41 and about 66 bits (preferably above 45), with shapes that emulate standardized curves. Each should have Frobenius conductor `f > 1`, so that its isogeny class has volcanoes with more than one level, as the root of an experiment that iterates over every curve in the isogeny class.

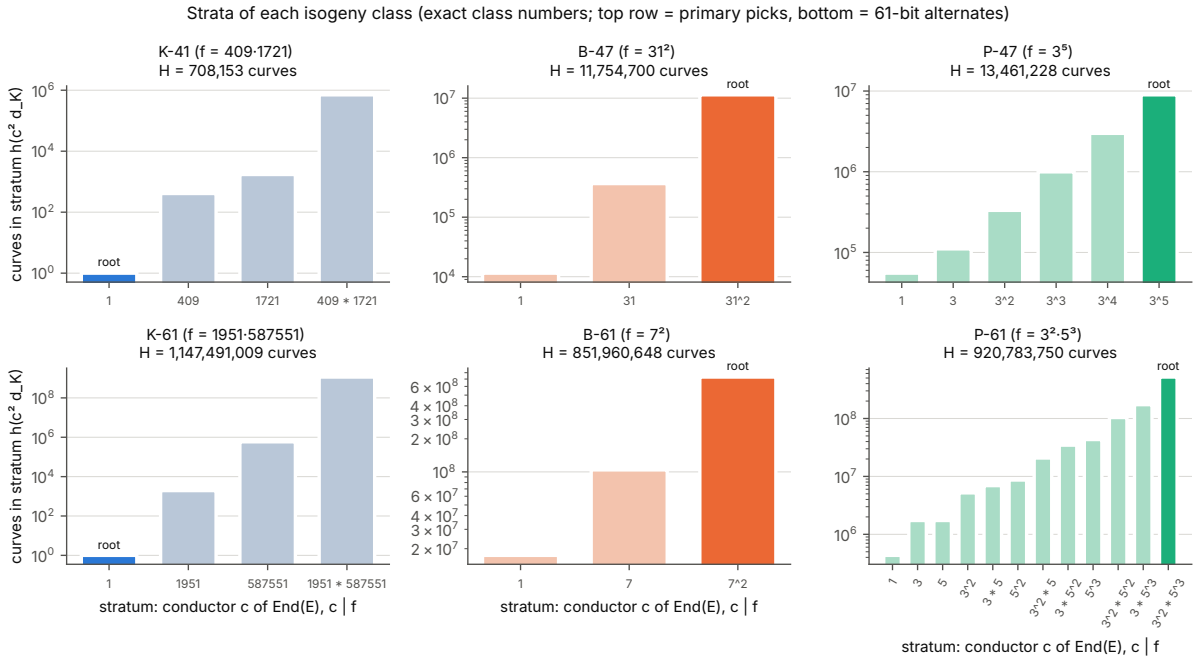
Picks

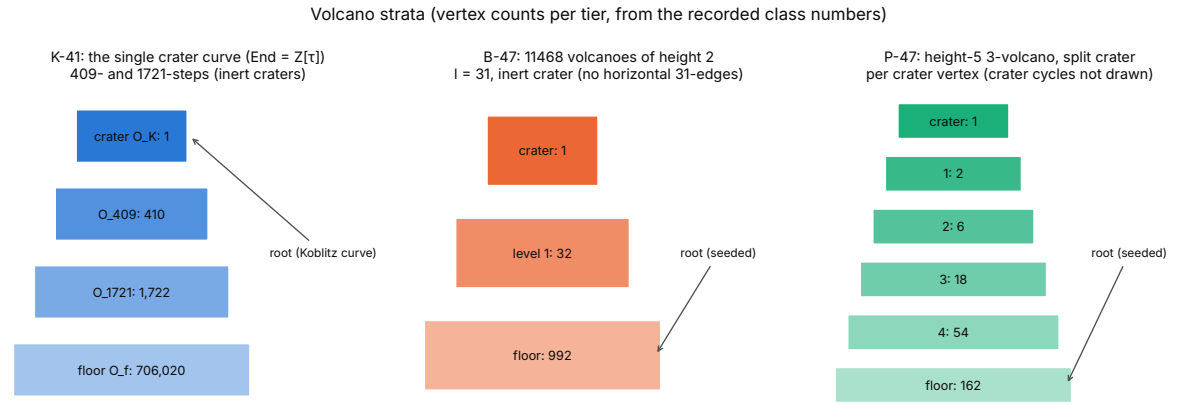
	Koblitz (K-shape)	Binary (B-shape)	Prime (P-shape)
ICV1 slug	`icv1-f2m41-tm2308219-7f48b14a`	`icv1-f2m47-t4276051-826c5755`	`icv1-fp47-tm8785349-43e36ce5`
Emulates	K-233/283/409/571 (`a = 0`, cofactor 4)	B-163…B-571 (`a = 1`, seeded `b`, cofactor 2)	P-192…P-521 (`a = −3`, seeded `b`, prime order)
Field	`GF(2^41)`, `x^41 + x^3 + 1`	`GF(2^47)`, `x^47 + x^5 + 1`	`GF(p)`, `p = 2^47 − 115`
Model	`y^2 + xy = x^3 + 1`	`y^2 + xy = x^3 + x^2 + b`, `b = 0x52e3db551eb3`	`y^2 = x^3 − 3x + b`, `b = 0xda8a554f2ba`
Seed	— (fixed by the family)	`isogeny-class-trio/2026-10-17/B-47/b/28948`	`isogeny-class-trio/2026-10-17/P-47/b/117828`
`#E`	`4 · 549756390943`	`2 · 70368742039639`	`140737497140563` (prime)
Prime subgroup	40 bits	46 bits	48 bits
Trace `t`	−2308219	4276051	−8785349
`d_K`	−7	−589770391	−8226516899
Conductor `f`	`409 · 1721`	`31^2`	`3^5`
Volcanoes	`l = 409`: height 1, inert; `l = 1721`: height 1, inert	`l = 31`: height 2, inert crater	`l = 3`: height 5, split crater
Strata (`c | f`)	4	3	6
Class size `H`	**708,153**	**11,754,700**	**13,461,228**
Root's position	crater (`End = Z[τ] = O_K`)	floor (`End = Z[π]`, conductor 961)	floor (`End = Z[π]`, conductor 243)
Generator	`(0x1deeba21c88, 0x17ccbb7d40e)` (registry, `EC1N41Ce0he09550ab560a`)	`(0xd6dc0d063d1, 0x23397c1623ca)`	`(0x4548698c68b3, 0x2b74b4fbfbc3)` (`EC1P47Cfph7e9290c297bc`)

Stratum sizes (curves with `End = O_c`, exact):

- K-41: `c = 1`: 1; `409`: 410; `1721`: 1,722; `409·1721`: 706,020.
- B-47: `c = 1`: 11,468; `31`: 366,976; `31^2`: 11,376,256.

- P-47: c = 1: 55,396; 3: 110,792; 9: 332,376; 27: 997,128; 81: 2,991,384; 243: 8,974,152.



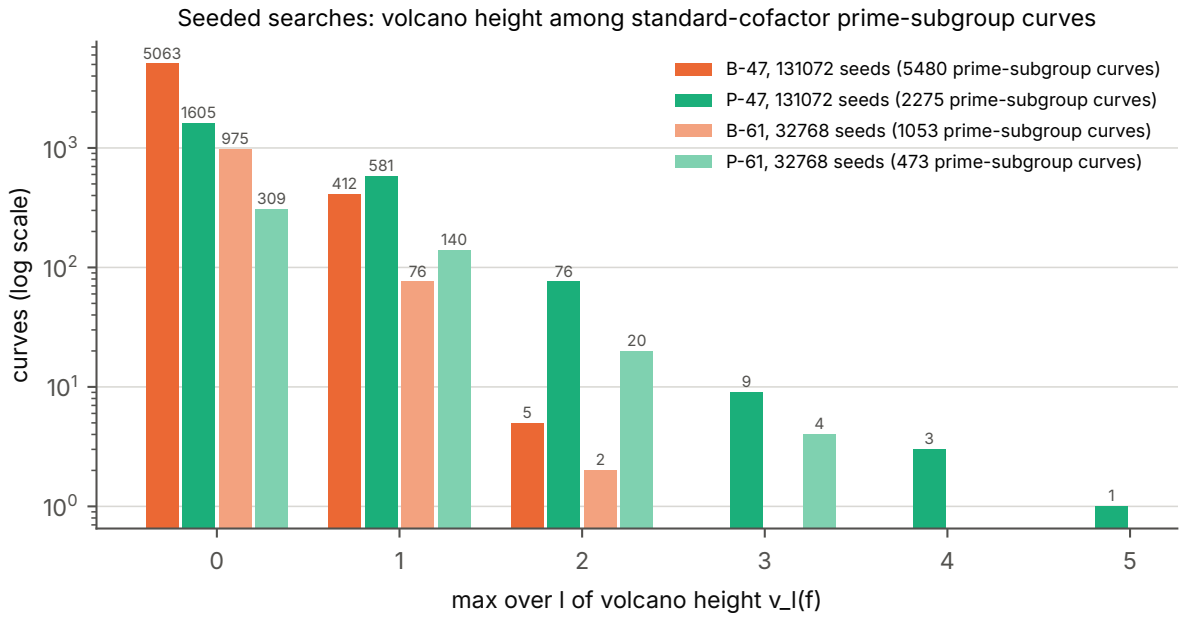


61-bit alternates (for a trio entirely above 45 bits)

	K-61	B-61	P-61
ICV1 slug	`icv1-f2m61-t158598901-ab42b6c5`	`icv1-f2m61-tm2657393969-b9ff0fb0`	`icv1-fp61-tm813574577-a3b71255`
Field	`x^61 + x^5 + x^2 + x + 1`	same	`p = 2^61 − 1` (as P-521 = `2^521 − 1`)
Model	`a = 0`, `b = 1`	`a = 1`, `b = 0x9c60f836d06fc8` (seed index 12079)	`a = −3`, `b = 0x54aa66b458b77ec` (seed index 11526)
Cofactor, subgroup	14156, 48 bits	2, 61 bits	1, 62 bits
`f`	`1951 · 587551`	`7^2` (split crater)	`3^2 · 5^3` (3 inert, 5 split)
Strata, `H`	4, 1,147,491,009	3, 851,960,648	12, 920,783,750
Root	crater	floor	floor of both volcanoes

Findings

1. **No prime-degree Koblitz curve has the standard cofactor between `2^42` and `2^66`** (derived, exhaustive). The prime degrees in range are 41, 43, 47, 53, 59 and 61; of the 12 curves only `K_0` over `GF(2^41)` has `#E = 4r` with `r` prime (`data/koblitz.jsonl`, figure). The K pick is therefore 41 bits, below the "preferably above 45" preference. Above 45 bits the best prime-degree option is `K_0` over `GF(2^61)` (cofactor 14156, 48-bit subgroup), listed as the alternate.
2. **A prime-degree Koblitz class has height-1 volcanoes at large `l` only** (derived; classical). `f = |U_m|`, the Lucas sequence of `τ^2 − μτ + 2`. For prime `m ≠ 7` every prime `l | U_m` has rank of apparition `m`, so `l ≡ ±1 (mod m)` and `l ≥ 2m − 1`; the table bears this out (409, 1721 ≡ −1, −1 mod 41; 1951, 587551 ≡ −1, −1 mod 61). The Koblitz curve is the only curve in its class with `End = O_K` (`h(−7) = 1`), so for every small `l` its isogeny graph is one vertex with loops: **the rest of the class is reachable only through the vertical steps of degree `l | f`.** For K-41 those are degree 409 and 1721, with kernel x-coordinates in `F_{q^204}` and `F_{q^215}` (computed: the order of `±t/2` in `F_l^*/±1`). For K-61 the second step has degree 587551 and kernel field `F_{q^58755}`, which is why K-61 is only the alternate.
3. **Binary curves over odd-degree fields have conductor primes `l ≡ ±1 (mod 8)` only** (derived; confirmed by the search). `l | f` needs `t^2 ≡ 2^(m+2) (mod l)`, i.e. 2 a square mod `l`. So 3- and 5-volcanoes never occur for B- or K-shape curves; the smallest possible vertical degree is 7. In the B-47 search every conductor prime was in {7, 17, 23, 31, 41, 47, 71, …} (`data/tally-B47.json`). For the same reason only 5 of 5,480 standard-cofactor curves have a height-2 volcano, against 89 of 2,275 for P-47.
4. **Both seeded roots sit on the floor** (proved by the tests below), as most curves do: the floor holds 96.8% of B-47's class and 66.7% of P-47's. Their crater representatives are 2 (B-47) and 5 (P-47) ascending isogenies away.



Method

- **Koblitz.** `#E(GF(2^m)) = 2^m + 1 − V_m` and `f = |U_m|` from the Lucas sequences of τ. The selector asserts `t^2 − 2^(m+2) = −7 U_m^2` for every row. The orders of K-41 and K-61 match their registry entries (`icv1-f2m41-tm2308219-7f48b14a`, `icv1-f2m61-t158598901-ab42b6c5`), and the native ICV1 producer reproduces their slugs.
- **B and P searches.** Coefficients come from SHA-256 of public seed strings, as in X9.62/FIPS 186 seeding. The selector counts points by BSGS over the Hasse window on seeded points, using `#E ≡ 2 (mod 4)` for B. A curve is kept only when the match is unique and `r = #E/h` is prime. `r > 4√q` then makes `#E` the only multiple of `r` in the window, so each kept order is proved. Not one count was ambiguous across the 327,680 curves.
- **Selection rule (frozen).** `f > 1`, `f` 61-smooth, a volcano of height ≥ 2. Rank by height, then number of strata, then lowest seed index.
- **Class invariants.** `Δ = t^2 − 4q` is factored to `f^2 d_K`. `h(d_K)` is computed exactly by counting reduced forms (`O(√|d_K|)`; cross-checked against brute force for every fundamental `|D| < 4000`). Each stratum's `h(c^2 d_K)` comes from the conductor formula, and `H = Σ_{c|f} h(c^2 d_K)` (Schoof 1987, Thm 4.6).
- **Root levels.**
  - **K:** `τ ∈ End(E)` gives `End = O_K` (crater).
  - **B:** for odd `l | f`, above the floor π acts on `E[l]` as the scalar `t/2`, so all `(l^2 − 1)/2` x-coordinates of `E[l]` lie in `F_{q^k}`, with `k` the order of `±t/2` in `F_l^*/±1`. On the floor `π − t/2` is a non-zero nilpotent, so only the eigenline's `(l − 1)/2` x-coordinates do. The test is therefore `deg gcd(ψ_l, x^(q^k) − x)` using characteristic-2 division polynomials. B-47 gives 15 of 480 (floor). Control: over `GF(2^7)` the Koblitz class has `f = 7`, and the test flags exactly the Koblitz curve as above the floor and the other 7 class members as floor (`floor_test_control_koblitz_m7`).
  - **P:** a 3-isogeny BFS (roots of `ψ_3`, Vélu) finds that the root has exactly one rational 3-isogeny. The repository walker independently agrees: `src/bin/isogeny_walk.rs`, with the same trace, `Δ`, a 3-volcano of depth 5, and one kernel-certified 3-edge from the root (`data/walker-class-P47.txt`). For P-61 the walker records one 3-edge and one 5-edge from the root, so the root is on both floors.
- **Tests.** `cargo test --release` in `selector/` runs 15 tests:

- BSGS against naive counts, in both characteristics;
- characteristic-2 3-isogenies against brute-force orders, plus their duals;
- division polynomials against rational torsion;
- class numbers against brute force;
- SHA-256 vectors;
- the floor-test control.

Status of each statement

Statement	Status
Orders, traces, `f`, `d_K` of all six curves	computed (native); order proved by `r > 4√q` (B, P) or exact formula (K)
`h(d_K)` and stratum sizes, `H`	computed exactly (native form count); formula classical
Root levels	B, P: proved by the tests above; P cross-checked by the walker; K: theorem
No standard-cofactor prime-degree Koblitz in 42–66	exhaustive enumeration
`l ≡ ±1 (mod m)` for Koblitz, `±1 (mod 8)` for binary	classical arguments, written above; consistent with all data
Seeded picks are optimal under the rule	within the frozen seed ranges only

No theorem in this report is new, so there is no Lean obligation (AGENTS.md, "Mathematical results should include Lean proofs"). Every structural fact used is classical and its argument is written out above.

What the whole-class experiment will need (not done here)

- **P-47** (13.5 M curves): the repository walker supports this field and all odd `l`, but it holds about 87 KB per curve (about 1.2 TB for this class). A full pass needs a compact class-group enumeration such as `p61_class` segments.
- **B-47** (11.8 M curves): there is no binary isogeny walker (`docs/isogeny-walk/README.md`: "Prime fields and odd l only").
  - It needs characteristic-2 `l`-isogenies: degree 31 for the vertical steps, and split primes (`(d_K/l) = 1`) for horizontal steps.
  - This selector already has `ψ_l` and 3-isogeny Vélu in characteristic 2.
  - 2-isogenies are horizontal here (2 splits, since `d_K ≡ 1 mod 8`).
- **K-41** (708 k curves): vertical isogenies of degree 409 and 1721 computed from kernels over `F_{q^204}` and `F_{q^215}` (or by evaluating `Φ_l`), then horizontal walks inside each stratum.

Reproduce

```
cd research/isogeny_class_trio_20261017/selector
cargo test --release && cargo build --release
B=target/release/isogeny-class-trio-selector
$B koblitz --min 41 --max 64                       > ../data/koblitz.jsonl
$B binary --m 47 --count 131072 --threads 4        > ../data/search-B47.jsonl 2> ../
data/search-B47.summary.json
$B prime --bits 47 --count 131072 --threads 4      > ../data/search-P47.jsonl 2> ../
data/search-P47.summary.json
$B binary --m 61 --count 32768 --threads 4         > ../data/search-B61.jsonl 2> ../
data/search-B61.summary.json
```

```
$B prime --bits 61 --count 32768 --threads 4       > ../data/search-P61.jsonl 2> ../
data/search-P61.summary.json
$B analyze-binary --m 41 --a2 0 --b 1 --koblitz    > ../data/pick-K41.json
$B analyze-binary --m 47 --a2 1 --b 0x52e3db551eb3 --seed isogeny-class-
trio/2026-10-17/B-47/b/28948 > ../data/pick-B47.json
$B analyze-prime --p 140737488355213 --b 0xda8a554f2ba --seed isogeny-class-
trio/2026-10-17/P-47/b/117828 > ../data/pick-P47.json
$B tally --in ../data/search-B47.jsonl             > ../data/tally-B47.json
```

Wall time on the 4-core session host: B-47 search 1 m 59 s, P-47 search 4 m 18 s, analyses under 5 s each. Figures: `python3 figures/plot.py` (plots recorded values only). The curves are registered in `docs/curves/registry.json` from `curves.json`.